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标题	High-yielding Pd2(dba)3 · C6H6-based four-fold Sonogashira coupling with selenophene-conjugated magnesium tetraethynylporphyrin for organic solar cells †
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摘要（原文节选）

A catalytic system using Pd2(dba)3 · (C6H6)/PPh3/CuI for Sonogashira coupling was demonstrated to synthesize a selenophene-conjugated magnesium tetraethynylporphyrin Mg-TEP-(Se-DPP)4 (2a). The catalytic system enabled four-fold cross-coupling of the four terminal alkynes of magnesium tetraethynylporphyrin with bromoselenophene-tethered diketopyrrolopyrroles (DPPs) to produce the desired star-shaped 2a in 80% yield. This molecule shows higher solubility in organic solvents, more efficient visible and near-infrared reg ⋯

数据表 · 表 1（5 行，含数值行 4 行）

Frontier orbital energies of Mg-TEP-(X-DPP-Ar)4 in solution as determined by electrochemical measurement and in solids as determined by photoelectron yield spectroscopy

Entry	Film	Solutiona	Solidc					
λ max [nm]	λ onset [nm]	λ max [nm]	λ onset [nm]	E ox 1/2 [V]	E red 1/2 [V]	HOMO [eV]	LUMO [eV]	E g [eV]
2a	620, 840	920	600, 794	866	0.47	− 1.11	− 5.27	− 3.69
2b	635, 848	950	617, 801	885	0.39	− 1.14	− 5.19	− 3.66
2c	650, 881	996	628, 816	905	0.32	− 1.25	− 5.12	− 3.55
3a	606, 826	867	587, 781	847	0.62	− 1.01	− 5.42	− 3.79

脚注（逐字）：aValues were determined by DPV. Measurements were performed in THF solution containing TBAPF6 (0.1 M) as a supporting electrolyte at 25 ° C with a scan rate of 100 mV s − 1. Glassy-carbon, platinum wire, and Ag/AgCl electrodes were used as the working, counter, and reference electrodes, respectively. T

正文对应句（逐字）：Based on the above considerations and existing challenges, we are interested in systematically exploring the effects of selenium substitution and end-capping with alkyl chains of thiophenes on the photovoltaic performance of SMs with D − (π − A)4 and D − (π − A − Ar)4 frameworks. In this work, we designed and

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